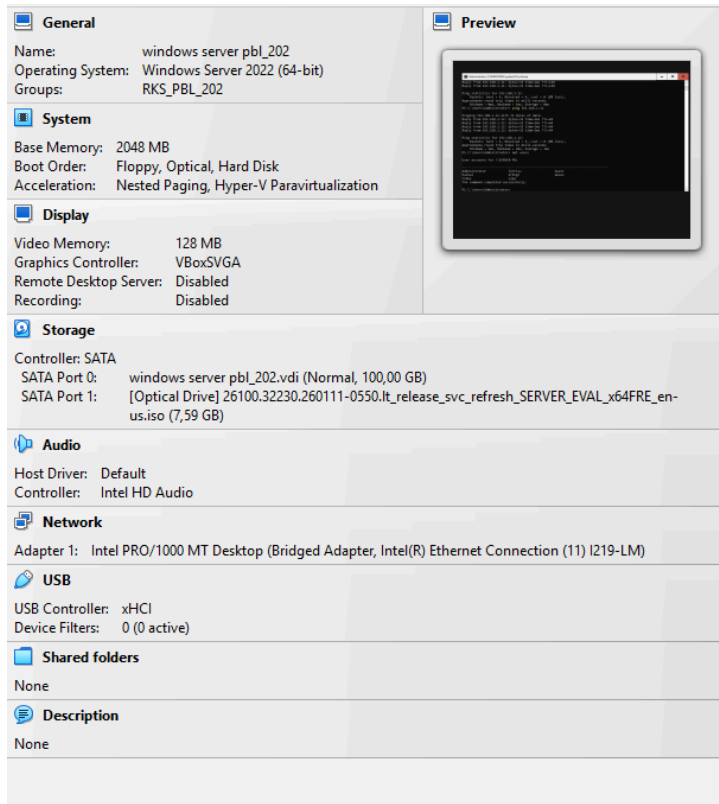
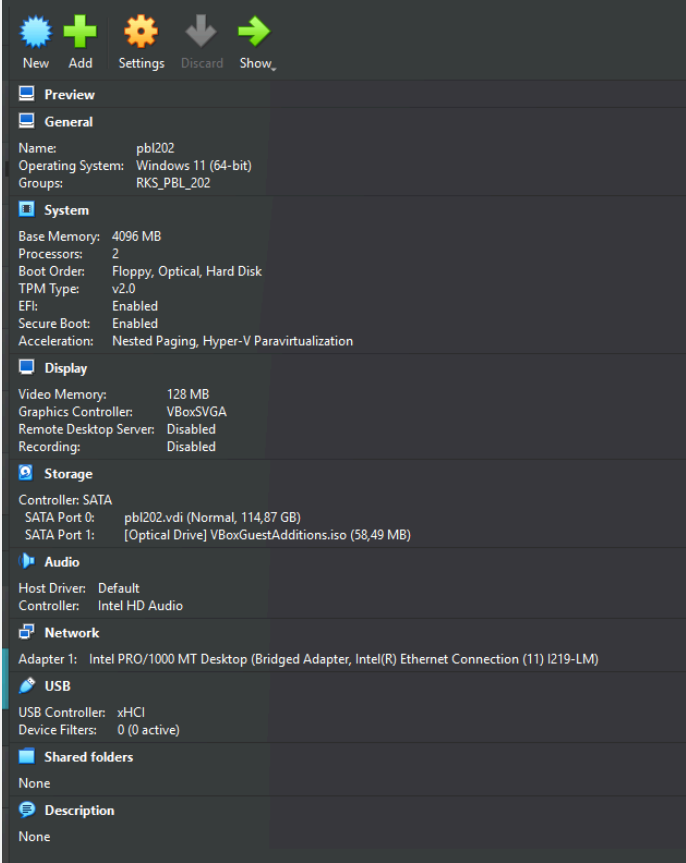
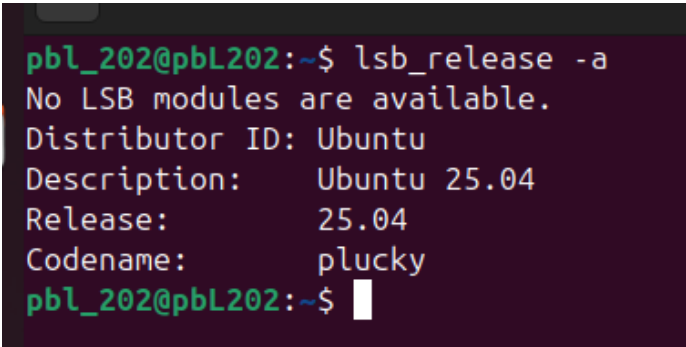
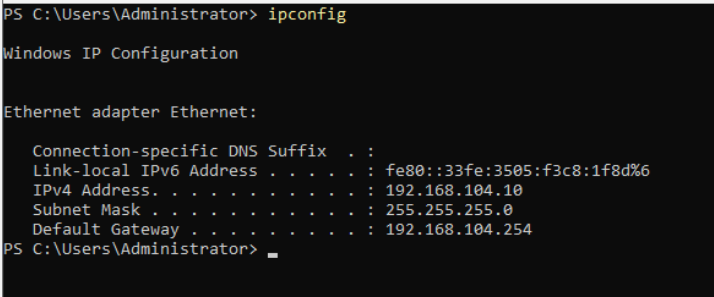


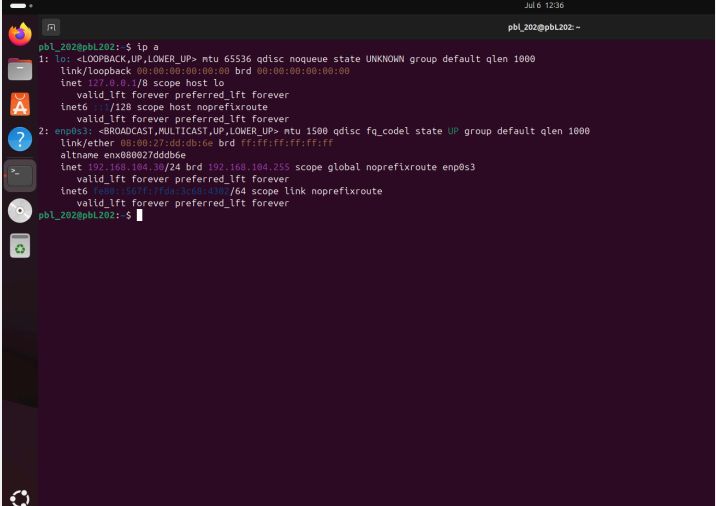
JOB SHEET PBL: RKS202 DASAR DASAR WINDOWS

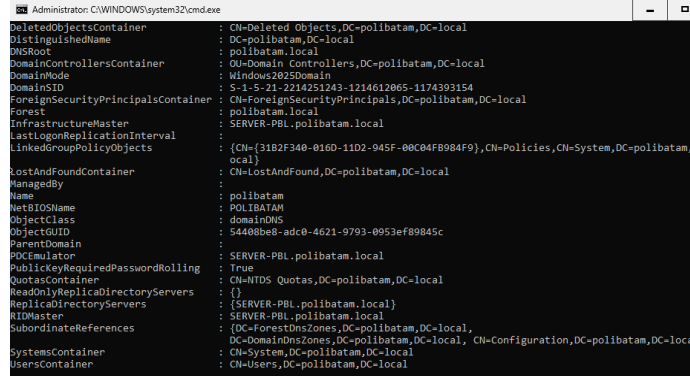
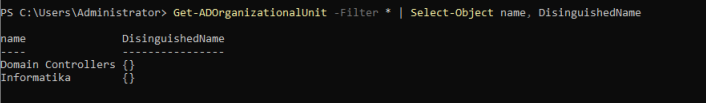
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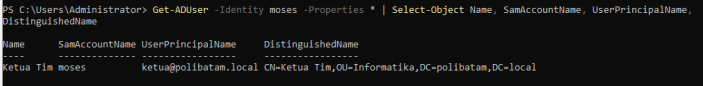
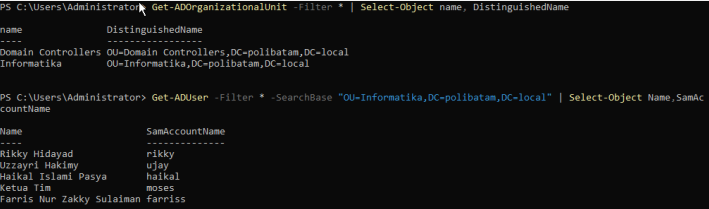
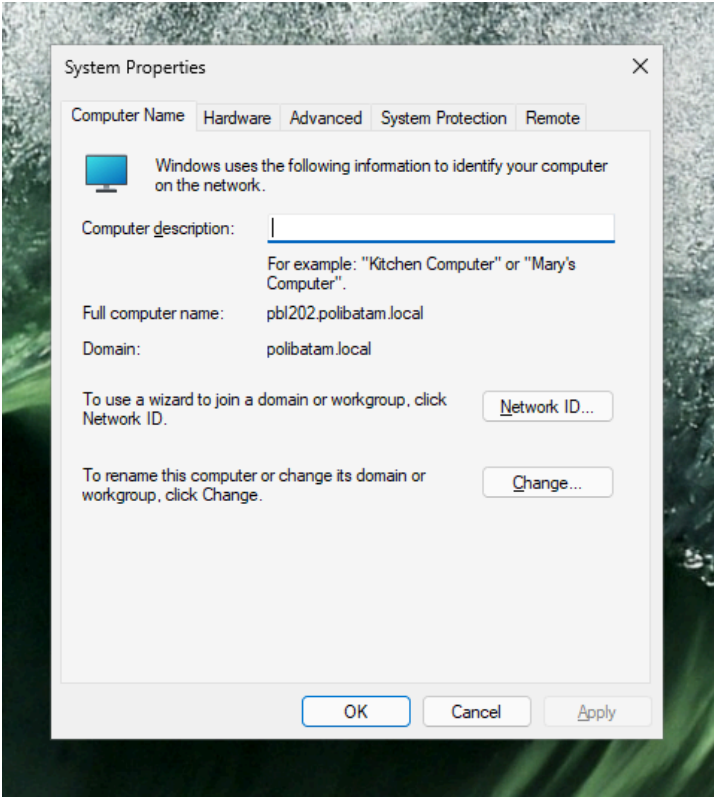
1. 4332511049 - Moses Hendika Purba
2. 4332511041 - Uzzayri Hakimy
3. 4332511040 - Haikal Islami Pasya
4. 4332511044 - Farris Nur Zakky Sulaiman
5. 4332511037 - Rikky Hidayad

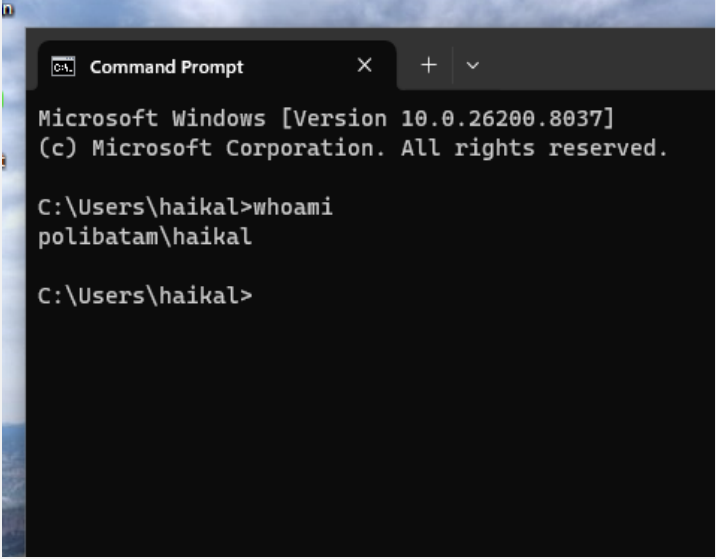
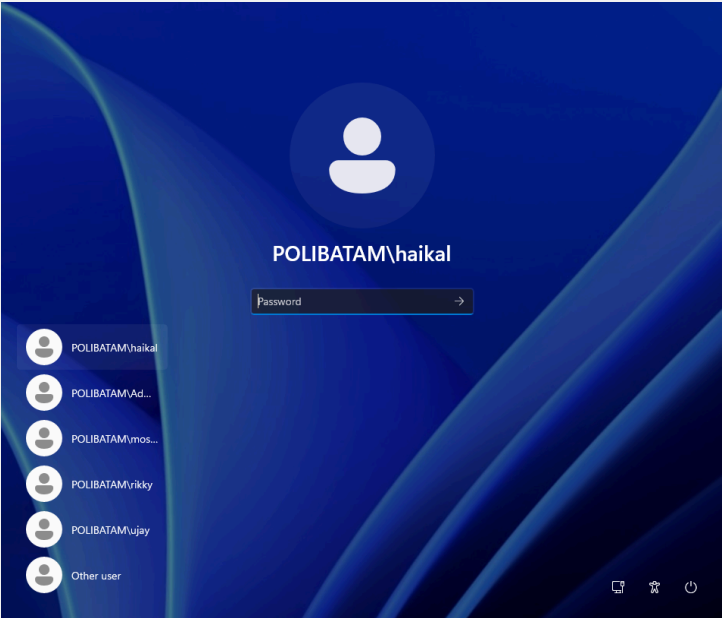
No	Kategori / Sub-Tugas	Keterangan	Screenshot	status
1 Desain Jaringan & Topologi – 10%				
1.1	Menyiapkan VM 1: Windows Server	Screenshot spesifikasi VM dari VMWare atau VirtualBox		☒

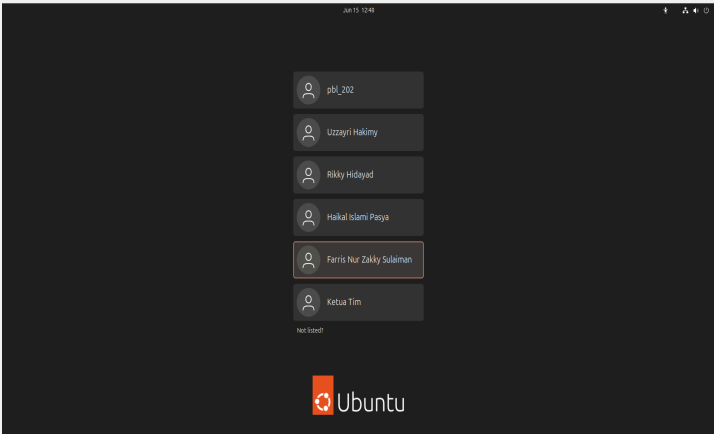
1.2 Menyiapkan VM 2: Windows 11 Client	Screenshot spesifikasi VM dari VMWare atau VirtualBox	 The screenshot shows the Virtual Machine settings for a VM named 'pbl202'. The 'General' tab is selected, showing the operating system as 'Windows 11 (64-bit)'. The 'System' tab shows 4096 MB of base memory and 2 processors. The 'Display' tab shows 128 MB of video memory. The 'Storage' tab shows a SATA controller with two ports: 'pbl202.vdi (Normal, 114.87 GB)' and '[Optical Drive] VBoxGuestAdditions.iso (58.49 MB)'. The 'Audio' tab shows the host driver as 'Default' and the controller as 'Intel HD Audio'. The 'Network' tab shows an 'Intel PRO/1000 MT Desktop' adapter. The 'USB' tab shows the controller as 'xHCI' and 0 active device filters. The 'Shared folders' and 'Description' tabs are also visible, both showing 'None'.	☒
1.3 Menyiapkan VM 3: Ubuntu Desktop	Screenshot terminal Ubuntu menampilka n versi OS	 <pre>pbl_202@pbl202:~\$ lsb_release -a No LSB modules are available. Distributor ID: Ubuntu Description: Ubuntu 25.04 Release: 25.04 Codename: plucky pbl_202@pbl202:~\$</pre>	☒
1.4 Konfigurasi IP Statis Windows Server	Hasil <i>output</i> command ipconfig	 <pre>PS C:\Users\Administrator> ipconfig Windows IP Configuration Ethernet adapter Ethernet: Connection-specific DNS Suffix . : Link-local IPv6 Address : fe80::33fe:3505:f3c8:1f8d%6 IPv4 Address. : 192.168.104.10 Subnet Mask : 255.255.255.0 Default Gateway : 192.168.104.254 PS C:\Users\Administrator></pre>	☒

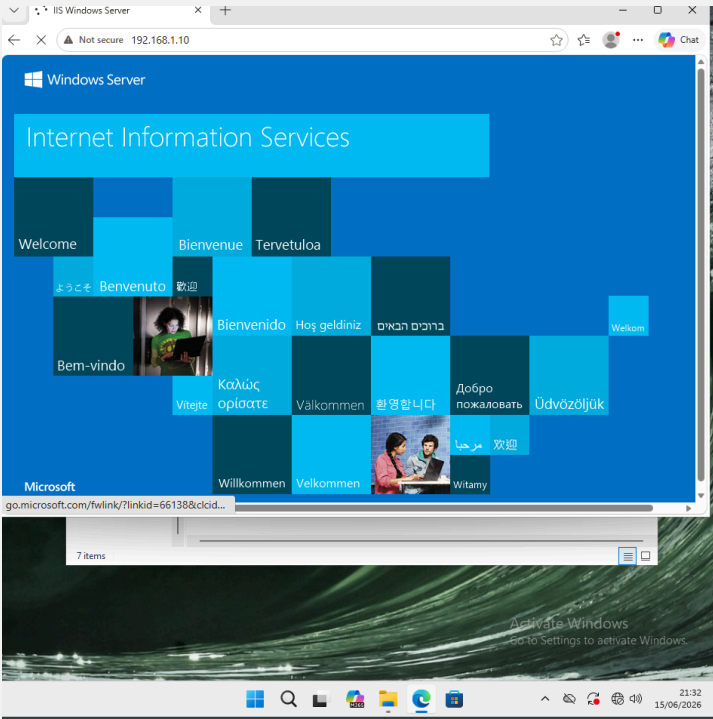
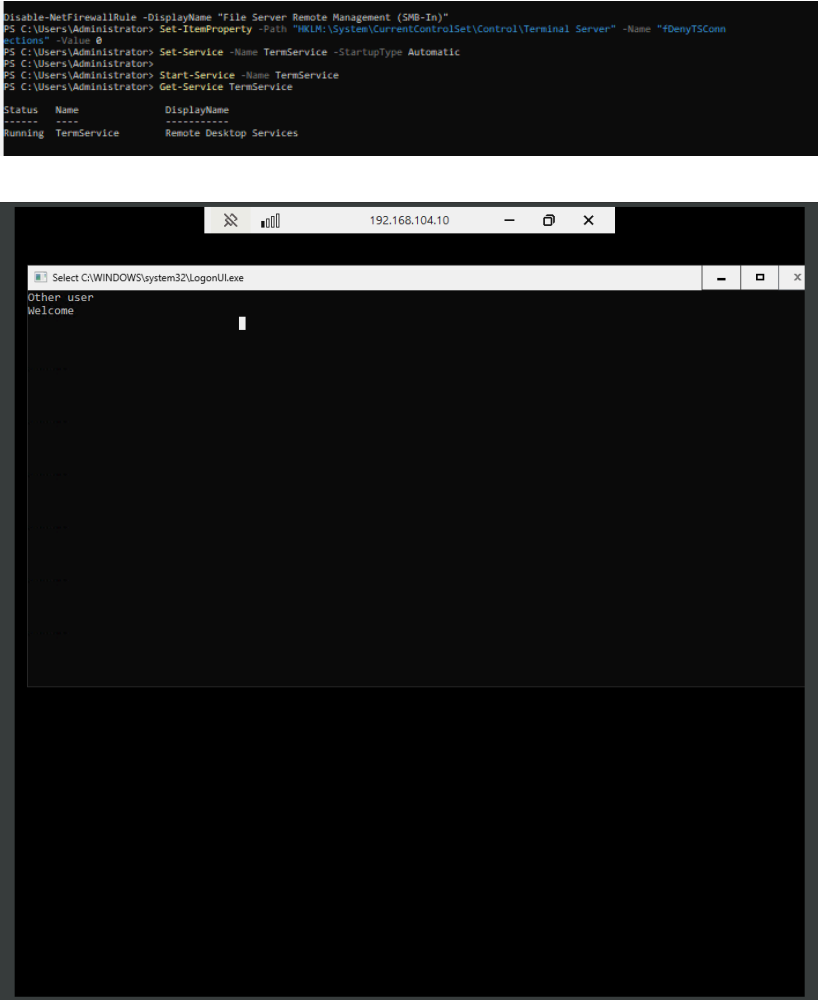
1.5 Konfigurasi IP Statik Windows 11 Client	Hasil <i>output</i> command ipconfig		<pre> C:\Users\PBL_202>ipconfig Windows IP Configuration Ethernet adapter Ethernet: Connection-specific DNS Suffix . : Link-local IPv6 Address : fe80::ee81:bed0:c23c:ceea%2 IPv4 Address. : 192.168.104.20 Subnet Mask : 255.255.255.0 Default Gateway : 192.168.104.254 C:\Users\PBL_202> </pre>	☒
1.6 Konfigurasi IP Statik Ubuntu Dekstop	Hasil <i>output</i> command ip a (Ubuntu)			☒
1.7 Pengujian koneksi antar VM	Screenshot hasil ping sukses dua arah antar Server dan kedua Client		Dari Windows Server Ke Windows 11 <pre> PS C:\Users\Administrator> ping 192.168.104.20 Pinging 192.168.104.20 with 32 bytes of data: Reply from 192.168.104.20: bytes=32 time=1ms TTL=128 Reply from 192.168.104.20: bytes=32 time=1ms TTL=128 Reply from 192.168.104.20: bytes=32 time=1ms TTL=128 Reply from 192.168.104.20: bytes=32 time=1ms TTL=128 Ping statistics for 192.168.104.20: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms PS C:\Users\Administrator> </pre> Dari Windows Server ke Ubuntu Dekstop <pre> PS C:\Users\Administrator> ping 192.168.104.30 Pinging 192.168.104.30 with 32 bytes of data: Reply from 192.168.104.30: bytes=32 time=2ms TTL=64 Reply from 192.168.104.30: bytes=32 time=2ms TTL=64 Reply from 192.168.104.30: bytes=32 time=2ms TTL=64 Reply from 192.168.104.30: bytes=32 time=2ms TTL=64 Ping statistics for 192.168.104.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 2ms, Average = 2ms PS C:\Users\Administrator> </pre> Dari Windows 11 ke Windows Server <pre> PS C:\Users\PBL_202\Desktop> ping 192.168.104.30 Pinging 192.168.104.30 with 32 bytes of data: Reply from 192.168.104.30: bytes=32 time=1ms TTL=64 Reply from 192.168.104.30: bytes=32 time=1ms TTL=64 Reply from 192.168.104.30: bytes=32 time=1ms TTL=64 Reply from 192.168.104.30: bytes=32 time=2ms TTL=64 Ping statistics for 192.168.104.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms </pre> Dari Windows 11 ke Ubuntu Server	☒

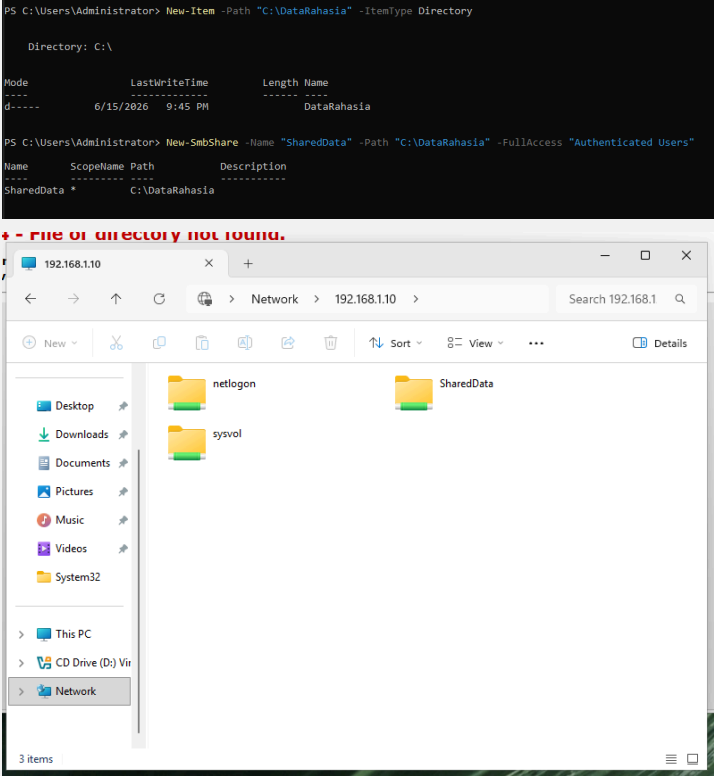
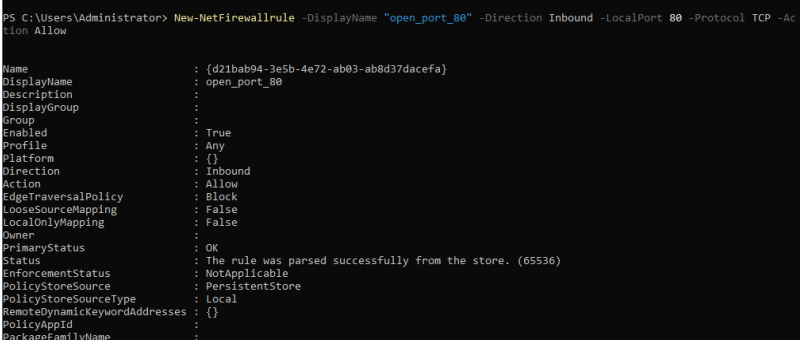
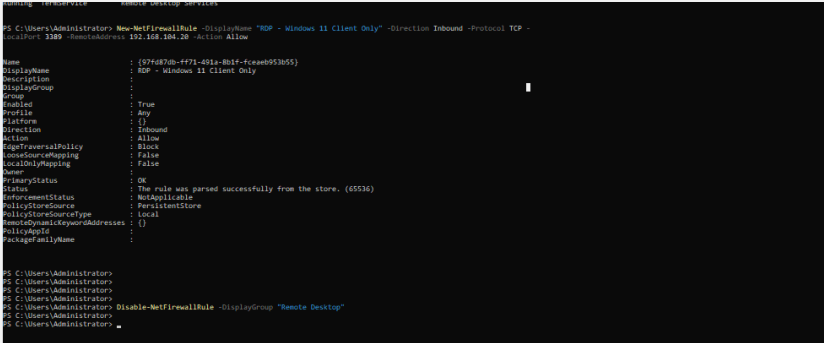
			<pre> PS C:\Users\PBL_202\Desktop> ping 192.168.104.30 Pinging 192.168.104.30 with 32 bytes of data: Reply from 192.168.104.30: bytes=32 time=1ms TTL=64 Reply from 192.168.104.30: bytes=32 time=1ms TTL=64 Reply from 192.168.104.30: bytes=32 time=1ms TTL=64 Reply from 192.168.104.30: bytes=32 time=2ms TTL=64 Ping statistics for 192.168.104.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms </pre> Dari Ubuntu Dekstop ke Windows Server <pre> pbl_202@pbl202:~\$ ping 192.168.104.10 PING 192.168.104.10 (192.168.104.10) 56(84) bytes of data: 64 bytes from 192.168.104.10: icmp_seq=1 ttl=128 time=1.48 ms 64 bytes from 192.168.104.10: icmp_seq=2 ttl=128 time=1.76 ms 64 bytes from 192.168.104.10: icmp_seq=3 ttl=128 time=1.54 ms 64 bytes from 192.168.104.10: icmp_seq=4 ttl=128 time=1.50 ms 64 bytes from 192.168.104.10: icmp_seq=5 ttl=128 time=1.72 ms 64 bytes from 192.168.104.10: icmp_seq=6 ttl=128 time=0.987 ms ^C --- 192.168.104.10 ping statistics --- 6 packets transmitted, 6 received, 0% packet loss, time 5013ms rtt min/avg/max/mdev = 0.987/1.498/1.760/0.252 ms pbl_202@pbl202:~\$ </pre> Dari Ubuntu Dekstop ke Windows 11 <pre> pbl_202@pbl202:~\$ ping 192.168.104.20 PING 192.168.104.20 (192.168.104.20) 56(84) bytes of data: 64 bytes from 192.168.104.20: icmp_seq=1 ttl=128 time=2.36 ms 64 bytes from 192.168.104.20: icmp_seq=2 ttl=128 time=0.886 ms 64 bytes from 192.168.104.20: icmp_seq=3 ttl=128 time=1.28 ms 64 bytes from 192.168.104.20: icmp_seq=4 ttl=128 time=1.84 ms 64 bytes from 192.168.104.20: icmp_seq=5 ttl=128 time=1.53 ms 64 bytes from 192.168.104.20: icmp_seq=6 ttl=128 time=1.32 ms 64 bytes from 192.168.104.20: icmp_seq=7 ttl=128 time=1.70 ms 64 bytes from 192.168.104.20: icmp_seq=8 ttl=128 time=1.22 ms 64 bytes from 192.168.104.20: icmp_seq=9 ttl=128 time=1.53 ms 64 bytes from 192.168.104.20: icmp_seq=10 ttl=128 time=0.951 ms 64 bytes from 192.168.104.20: icmp_seq=11 ttl=128 time=1.26 ms 64 bytes from 192.168.104.20: icmp_seq=12 ttl=128 time=1.01 ms 64 bytes from 192.168.104.20: icmp_seq=13 ttl=128 time=1.14 ms 64 bytes from 192.168.104.20: icmp_seq=14 ttl=128 time=0.844 ms 64 bytes from 192.168.104.20: icmp_seq=15 ttl=128 time=1.24 ms ^C --- 192.168.104.20 ping statistics --- 15 packets transmitted, 15 received, 0% packet loss, time 14108ms rtt min/avg/max/mdev = 0.844/1.340/2.360/0.390 ms </pre>	
2 Manajemen Identitas Terpusat – 30%				
2.1 Instalasi AD DS (polibatam.local)	Screenshot Dashboard Server Manager / Get-ADDomain di PowerShell		 <pre> Administrator C:\WINDOWS\system32\cmd.exe DeletedObjectsContainer : CN=Deleted Objects,DC=polibatam,DC=local DistinguishedName : DC=polibatam,DC=local DNSRoot : polibatam.local DomainControllersContainer : OU=Domain Controllers,DC=polibatam,DC=local DomainMode : Windows2025Domain DomainSID : S-1-5-21-2214251243-1214612865-1174393154 ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=polibatam,DC=local Forest : polibatam.local InfrastructureMaster : SERVER-PBL.polibatam.local LastLogonReplicationInterval : LinkedGroupPolicyObjects : {CN={31B2F340-8160-1102-945F-00C04FB984F9},CN=Policies,CN=System,DC=polibatam,DC=local} LostAndFoundContainer : CN=LostAndFound,DC=polibatam,DC=local ManagedBy : Name : polibatam NetBIOSName : POLIBATAM ObjectClass : domainDNS ObjectGUID : 54488be8-adc0-4621-9793-0953ef89845c ParentDomain : PDCemulator : SERVER-PBL.polibatam.local PublicKeyRequiredPasswordRolling : True QuotasContainer : CN=NTDS Quotas,DC=polibatam,DC=local ReadOnlyReplicaDirectoryServers : {} ReplicaDirectoryServers : {SERVER-PBL.polibatam.local} RIDMaster : SERVER-PBL.polibatam.local SubordinateReferences : {DC=ForestDnsZones,DC=polibatam,DC=local, CN=Configuration,DC=polibatam,DC=local, DC=DomainDnsZones,DC=polibatam,DC=local, CN=System,DC=polibatam,DC=local} SystemsContainer : CN=System,DC=polibatam,DC=local UsersContainer : CN=Users,DC=polibatam,DC=local </pre>	☒
2.2 Membuat Struktur OU	Screenshot konsol Active Directory Users and Computers (ADUC)		 <pre> PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * Select-Object name, DistinguishedName ----- name DistinguishedName ---- Domain Controllers {} Informatika {} </pre>	☒

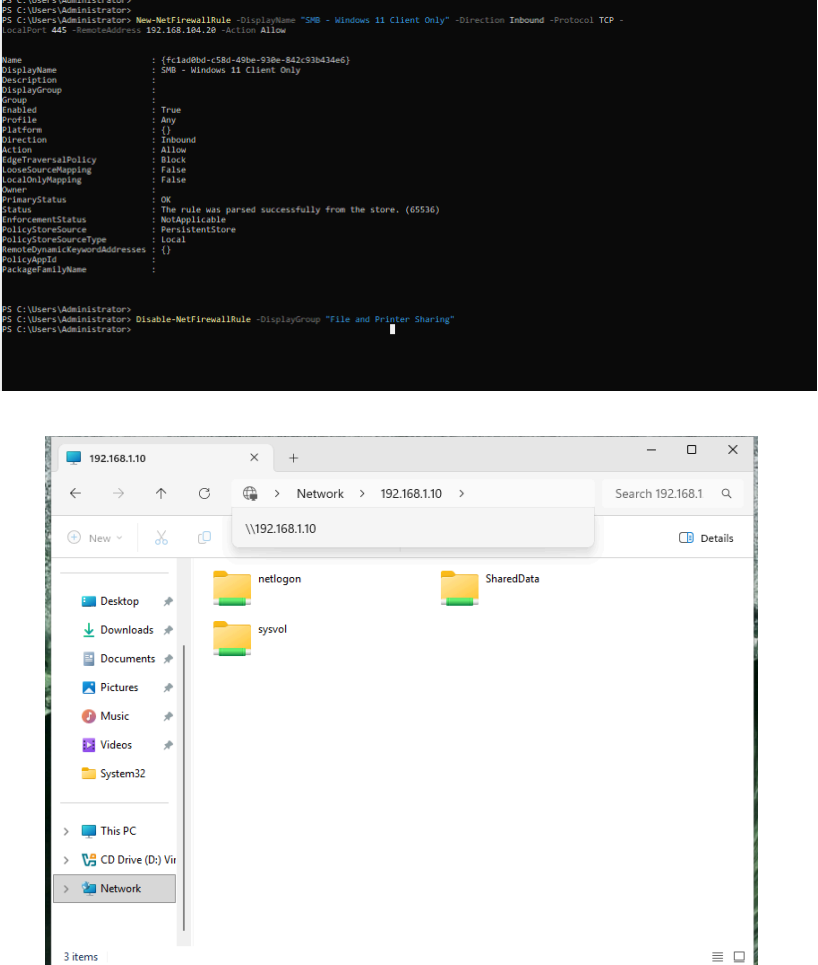
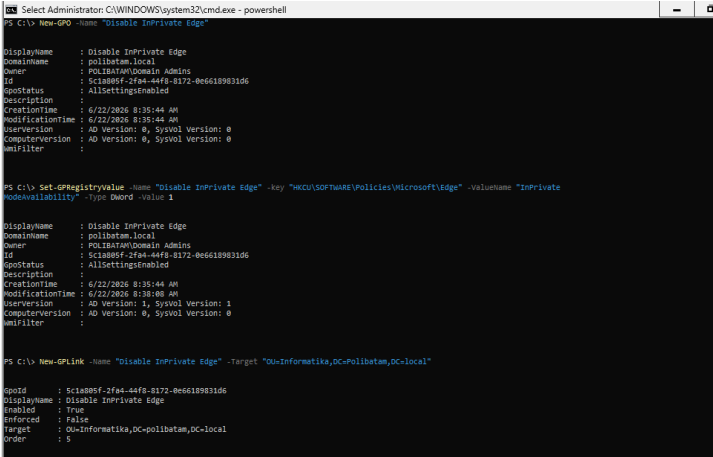
2.3 Akun Pengguna: Ketua Tim	Screenshot detail akun Ketua Tim di dalam OU yang sesuai	 <pre>PS C:\Users\Administrator> Get-ADUser -Identity moses -Properties * Select-Object Name, SamAccountName, UserPrincipalName, DistinguishedName ----- Name SamAccountName UserPrincipalName DistinguishedName ----- Ketua Tim moses ketua@polibatam.local CN=Ketua Tim,OU-Informatika,DC-polibatam,DC-local</pre>	☒
2.4 Akun Pengguna: Anggota Tim	Screenshot seluruh daftar akun Anggota Tim di dalam OU	 <pre>PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * Select-Object name, DistinguishedName ----- Name DistinguishedName ----- Domain Controllers OU=Domain Controllers,DC-polibatam,DC-local Informatika OU=Informatika,DC-polibatam,DC-local PS C:\Users\Administrator> Get-ADUser -Filter * -SearchBase "OU=Informatika,DC-polibatam,DC-local" Select-Object Name,SamAccountName ----- Name SamAccountName ----- Rikky Hidayad rikky Rizayri Hakimy rjay Haikal Islami Pasya haikal Ketua Tim moses Farris Nur Zakky Sulaiman Farris</pre>	☒
2.5 Join Domain Windows 11 Client	Screenshot menu <i>System Properties</i> Windows 11 menunjukkan status Domain		☒

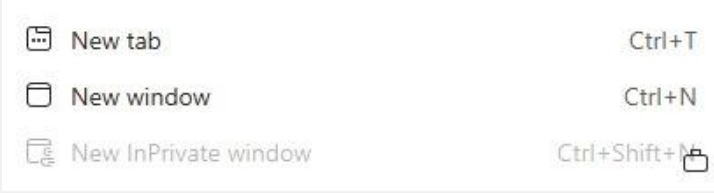
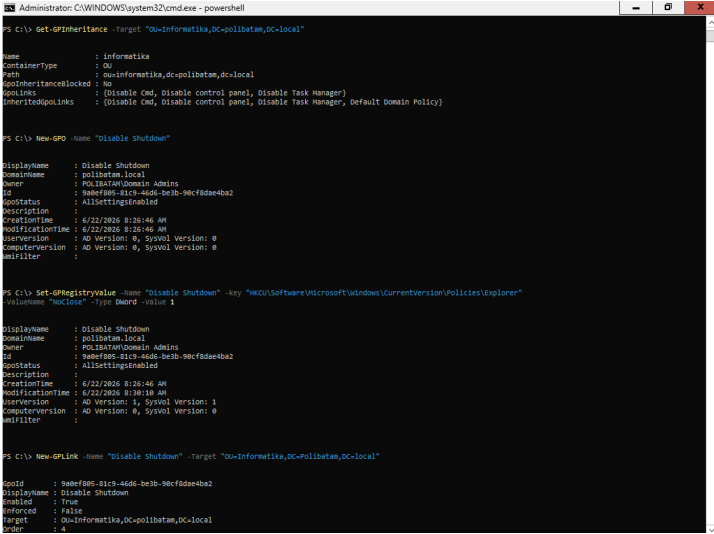
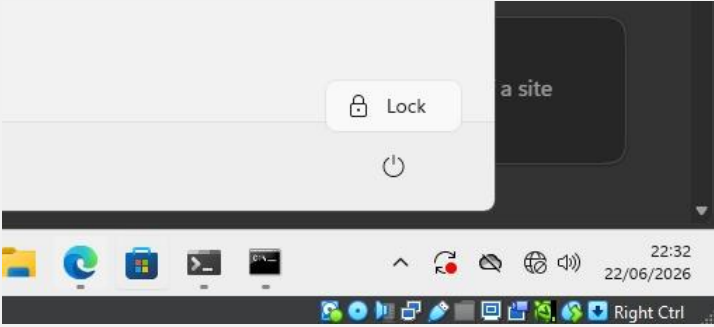
2.6 Pengujian Login Windows 11	Screenshot layar Desktop/Lock Screen dengan user AD yang sukses login	 Command Prompt window showing the output of the 'whoami' command. The text displayed is: Microsoft Windows [Version 10.0.26200.8037] (c) Microsoft Corporation. All rights reserved. C:\Users\haikal>whoami polibatam\haikal C:\Users\haikal>  Windows 11 login screen for user POLIBATAM\haikal. The screen shows a large user icon and the text 'POLIBATAM\haikal'. Below it is a password field labeled 'Password' with a right arrow button. On the left side, there is a list of user accounts: POLIBATAM\haikal, POLIBATAM\Ad..., POLIBATAM\mos..., POLIBATAM\riicky, POLIBATAM\ujay, and Other user. The background is a blue abstract design.
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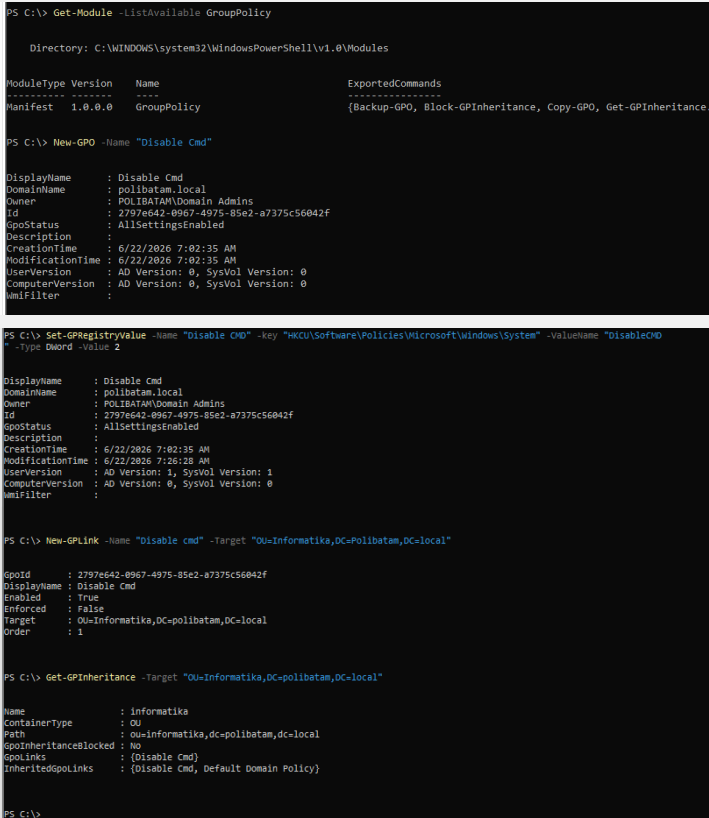
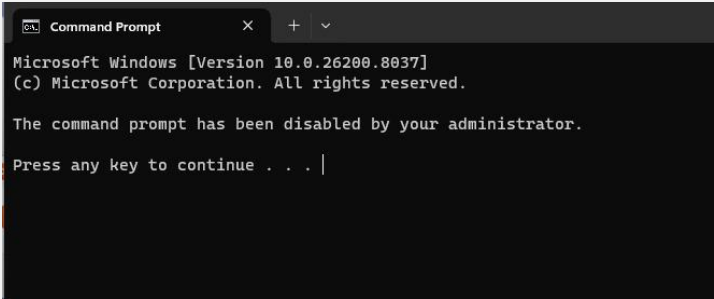
	2.7 Join Domain Ubuntu Desktop	Screenshot output perintah realm list atau file konfigurasi SSSD	<pre> pbl_202@pbl202:~\$ realm list polibatam.local type: kerberos realm-name: POLIBATAM.LOCAL domain-name: polibatam.local configured: kerberos-member server-software: active-directory client-software: sssd required-package: sssd-tools required-package: sssd required-package: libnss-sss required-package: libpam-sss required-package: adcli required-package: samba-common-bin login-formats: %U@polibatam.local login-policy: allow-realm-logins pbl_202@pbl202:~\$ </pre>	☒
	2.8 Pengujian Login Ubuntu Desktop	Screenshot layar Desktop/Lock Screen dengan user AD yang sukses login		☒
3 Keamanan Layanan & Firewall – 10%				

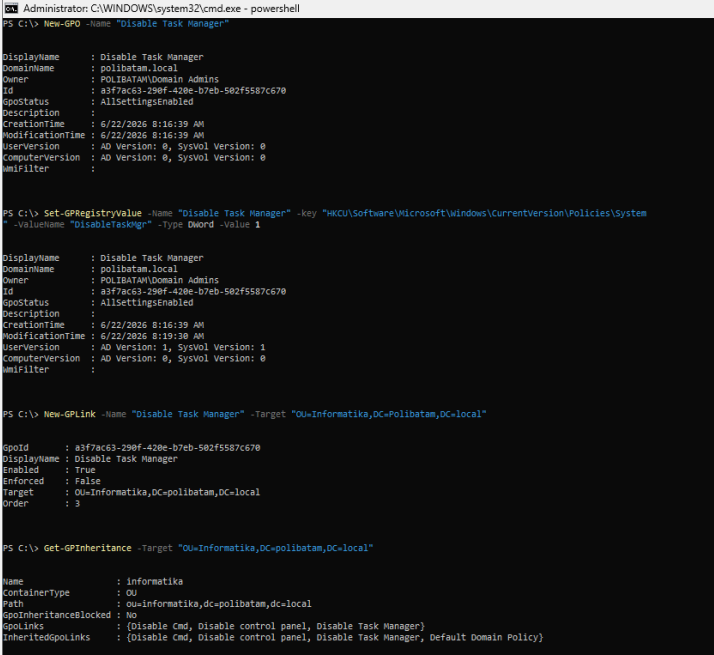
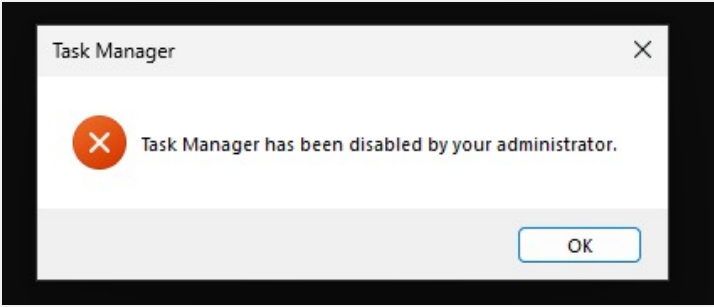
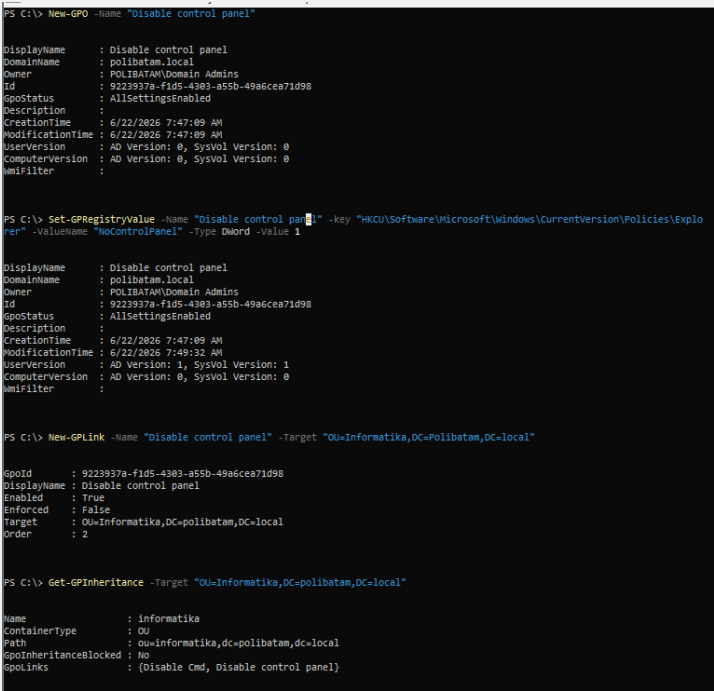
3.1 Instalasi Web Server (IIS/Nginx/Apache)	Screenshot halaman default web server saat diakses dari browser Client		☒
3.2 Mengaktifkan Fitur RDP	Screenshot pengaturan Remote Desktop aktif di Windows Server	 <pre>Disable-NetFirewallRule -DisplayName "File Server Remote Management (SMB-In)" PS C:\Users\Administrator> Set-ItemProperty -Path "HKLM:\System\CurrentControlSet\Control\Terminal Server" -Name "fDenyTSConnections" -Value 0 PS C:\Users\Administrator> Set-Service -Name TermService -StartupType Automatic PS C:\Users\Administrator> Start-Service -Name TermService PS C:\Users\Administrator> Get-Service TermService Status Name DisplayName ----- Running TermService Remote Desktop Services</pre>	☒

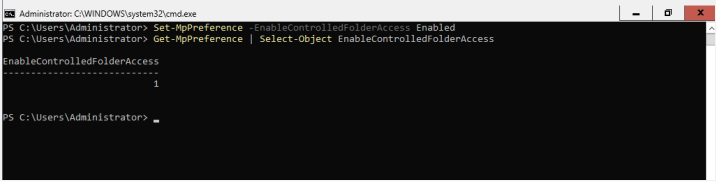
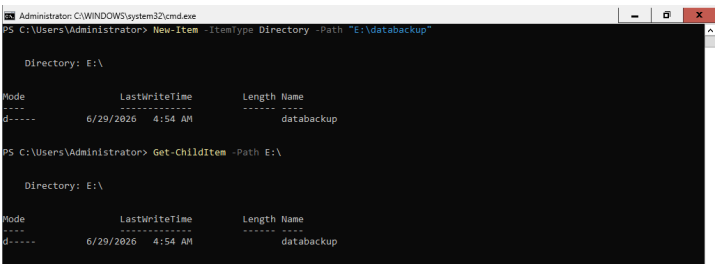
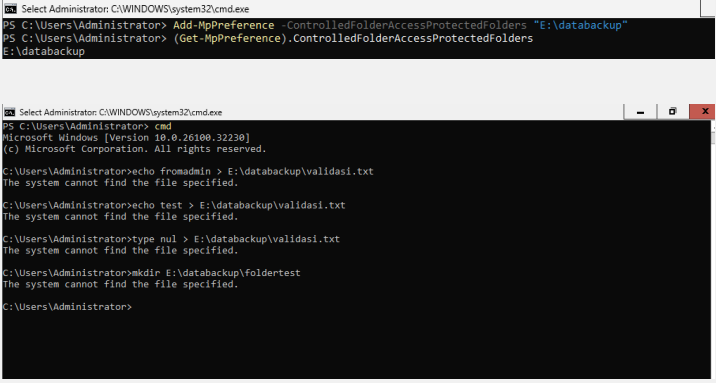
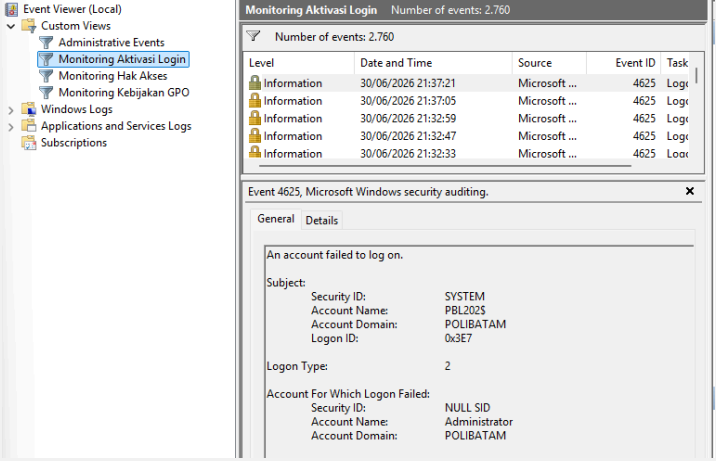
3.3 Mengaktifkan SMB (File Sharing)	Screenshot folder yang dibagikan (Shared Folder) di Windows Explorer	 The screenshot shows two parts. The top part is a PowerShell terminal window with the following commands and output: <pre>PS C:\Users\Administrator> New-Item -Path "C:\DataRahasia" -ItemType Directory Directory: C:\ Mode LastWriteTime Length Name ----- d----- 6/15/2026 9:45 PM DataRahasia PS C:\Users\Administrator> New-SmbShare -Name "SharedData" -Path "C:\DataRahasia" -FullAccess "Authenticated Users" Name ScopeName Path Description ----- SharedData * C:\DataRahasia</pre> The bottom part is a screenshot of Windows Explorer showing the network path 192.168.1.10. The left sidebar shows 'Network' selected. The main pane displays three folders: 'netlogon', 'sysvol', and 'SharedData'.	☒
3.4 Rule Firewall: HTTP Port 80	Screenshot <i>Inbound Rule</i> HTTP terbuka untuk umum di Windows Firewall	 The screenshot shows a PowerShell terminal window with the following command and output: <pre>PS C:\Users\Administrator> New-NetFirewallRule -DisplayName "open_port_80" -Direction Inbound -LocalPort 80 -Protocol TCP -Action Allow Name : {d21bab94-3e5b-4e72-ab03-ab8d37dacefa} DisplayName : open_port_80 Description : DisplayGroup : Group : Enabled : True Profile : Any Platform : {} Direction : Inbound Action : Allow EdgeTraversalPolicy : Block LooseSourceMapping : False LocalOnlyMapping : False Owner : PrimaryStatus : OK Status : The rule was parsed successfully from the store. (65536) EnforcementStatus : NotApplicable PolicyStoreSource : PersistentStore PolicyStoreSourceType : Local RemoteDynamicKeywordAddresses : {} PolicyAppId : PackageFamilyName :</pre>	☒
3.5 Rule Firewall: RDP Port 3389	Screenshot <i>Inbound Rule</i> RDP dengan <i>Scope</i> IP khusus milik Windows 11	 The screenshot shows a PowerShell terminal window with the following commands and output: <pre>PS C:\Users\Administrator> New-NetFirewallRule -DisplayName "RDP - Windows 11 Client Only" -Direction Inbound -Protocol TCP -LocalPort 3389 -RemoteAddress 192.168.104.20 -Action Allow Name : {57d87db-f771-401a-b01f-fc9aeb95b055} DisplayName : RDP - Windows 11 Client Only Description : DisplayGroup : Group : Enabled : True Profile : Any Platform : {} Direction : Inbound Action : Allow EdgeTraversalPolicy : Block LooseSourceMapping : False LocalOnlyMapping : False Owner : PrimaryStatus : OK Status : The rule was parsed successfully from the store. (65536) EnforcementStatus : NotApplicable PolicyStoreSource : PersistentStore PolicyStoreSourceType : Local RemoteDynamicKeywordAddresses : {} PolicyAppId : PackageFamilyName :</pre> Below the output, several commands are entered to manage the rule: <pre>PS C:\Users\Administrator> PS C:\Users\Administrator> PS C:\Users\Administrator> PS C:\Users\Administrator> Disable-NetFirewallRule -DisplayGroup "Remote Desktop" PS C:\Users\Administrator> PS C:\Users\Administrator></pre>	☒

3.6 Rule Firewall: SMB Port 445	Screenshot <i>Inbound Rule</i> SMB dengan <i>Scope</i> IP khusus milik Windows 11		☒ ☐
4 Kebijakan Keamanan (GPO) – 10%			
4.1 Konfigurasi kebijakan GPO 1 - Moses Hendika Purba	Screenshot konfigurasi GPO di <i>Group Policy Management Editor</i>		☒

4.2 Hasil GPO 1 di Windows Client - Moses Hendika Purba	Screenshot bukti kebijakan diterapkan di sisi Windows Client (misal: hasil gpresult)		☒
4.3 Kebijakan GPO 2 - Rikky Hidayad	Screenshot konfigurasi GPO di Group Policy Management Editor		☒
4.4 Hasil GPO 2 di Windows Client - Rikky Hidayad	Screenshot bukti kebijakan diterapkan di sisi Windows Client (misal: hasil gpresult)		☒

4.5 Kebijakan GPO 3 - Uzzayri Hakimy	Screenshot konfigurasi GPO di <i>Group Policy Management Editor</i>	 <pre> PS C:\> Get-Module -ListAvailable GroupPolicy Directory: C:\WINDOWS\system32\WindowsPowerShell\v1.0\Modules ModuleType Version Name ExportedCommands ----- Manifest 1.0.0.0 GroupPolicy {Backup-GPO, Block-GPInheritance, Copy-GPO, Get-GPInheritance, ...} PS C:\> New-GPO -Name "Disable Cmd" DisplayName : Disable Cmd DomainName : polibatam.local Owner : POLIBATAM\Domain Admins Id : 2797e642-0967-4975-85e2-a7375c56042f GpoStatus : AllSettingsEnabled Description : CreationTime : 6/22/2026 7:02:35 AM ModificationTime : 6/22/2026 7:02:35 AM UserVersion : AD Version: 0, SysVol Version: 0 ComputerVersion : AD Version: 0, SysVol Version: 0 WmiFilter : PS C:\> Set-GPRegistryvalue -Name "Disable Cmd" -key "HKCU\Software\Policies\Microsoft\Windows\System" -valueName "DisableCmd" -Type Dword -value 2 DisplayName : Disable Cmd DomainName : polibatam.local Owner : POLIBATAM\Domain Admins Id : 2797e642-0967-4975-85e2-a7375c56042f GpoStatus : AllSettingsEnabled Description : CreationTime : 6/22/2026 7:02:35 AM ModificationTime : 6/22/2026 7:26:28 AM UserVersion : AD Version: 1, SysVol Version: 1 ComputerVersion : AD Version: 0, SysVol Version: 0 WmiFilter : PS C:\> New-GPLink -Name "Disable Cmd" -Target "OU=Informatika,DC=polibatam,DC=local" GpoId : 2797e642-0967-4975-85e2-a7375c56042f DisplayName : Disable Cmd Enabled : True Enforced : False Target : OU=Informatika,DC=polibatam,DC=local Order : 1 PS C:\> Get-GPInheritance -Target "OU=Informatika,DC=polibatam,DC=local" Name : Informatika ContainerType : OU Path : OU=Informatika,DC=polibatam,DC=local GpoInheritanceBlocked : No GpoLinks : (Disable Cmd) InheritedGpoLinks : (Disable Cmd, Default Domain Policy) PS C:\> </pre>	☒
4.6 Hasil GPO 3 di Windows Client - Uzzayri Hakimy	Screenshot bukti kebijakan diterapkan di sisi Windows Client (misal: hasil gpresult)	 <pre> Command Prompt Microsoft Windows [Version 10.0.26200.8037] (c) Microsoft Corporation. All rights reserved. The command prompt has been disabled by your administrator. Press any key to continue . . . </pre>	☒

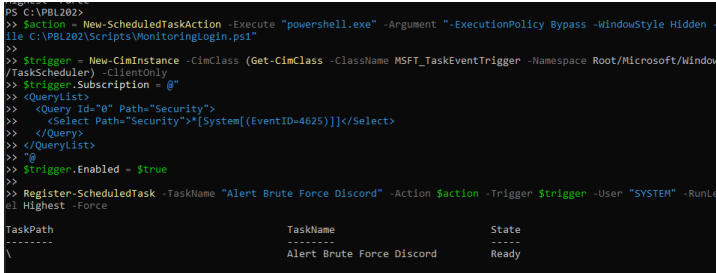
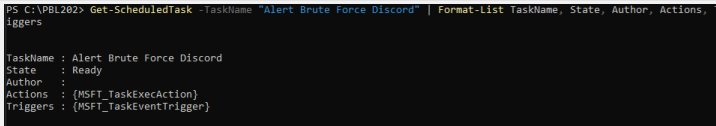
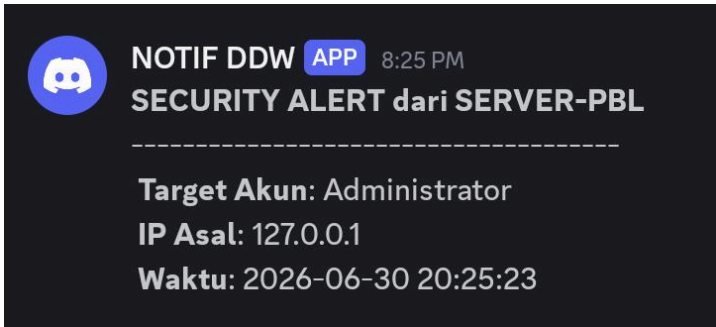
4.7 Kebijakan GPO 4 - Farris Nurzakky Sulaiman	Screenshot konfigurasi GPO di <i>Group Policy Management Editor</i>		☒
4.8 Hasil GPO 4 di Windows Client - Farris Nurzakky Sulaima	Screenshot bukti kebijakan diterapkan di sisi Windows Client (misal: hasil gpresult)		☒
4.9 Kebijakan GPO 5 - Haikal Islami Pasya	Screenshot konfigurasi GPO di <i>Group Policy Management Editor</i>		☒

5.2 Aktifkan <i>Controlled Folder Access</i>	Screenshot status <i>On</i> pada menu Windows Security di Windows Server		☒
5.3 Membuat folder databackup	Screenshot direktori D:\databackup di Windows Explorer		☒
5.4 Proteksi Ransomware folder	Screenshot daftar <i>Protected Folders</i> yang memuat path D:\databackup		☒
6 Audit & Monitoring Log – 5%			
6.1 Membuat <i>Custom View</i> Event Viewer	Screenshot panel Event Viewer yang menampilkan nama <i>Custom View</i> baru		☒

6.2 Filter Log: Aktivasi Login	Screenshot properti filter (Event ID 4624/4625) dan hasil lognya	Custom View Properties Filter XML Logged: Any time Event level: ☒ Critical ☒ Warning ☒ Verbose ☒ Error ☒ Information ☒ By log Event logs: Security ☐ By source Event sources: Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76 4624,4625 Task category: Keywords: User: <All Users> Computer(s): <All Computers> Clear OK Cancel Number of events: 2,760 <table><tr><th>Level</th><th>Date and Time</th><th>Source</th><th>Event ID</th><th>Task Categ...</th></tr><tr><td>Information</td><td>30/06/2026 21:37:21</td><td>Microsoft ...</td><td>4625</td><td>Logon</td></tr><tr><td>Information</td><td>30/06/2026 21:37:05</td><td>Microsoft ...</td><td>4625</td><td>Logon</td></tr><tr><td>Information</td><td>30/06/2026 21:32:59</td><td>Microsoft ...</td><td>4625</td><td>Logon</td></tr><tr><td>Information</td><td>30/06/2026 21:32:47</td><td>Microsoft ...</td><td>4625</td><td>Logon</td></tr><tr><td>Information</td><td>30/06/2026 21:32:33</td><td>Microsoft ...</td><td>4625</td><td>Logon</td></tr><tr><td>Information</td><td>30/06/2026 21:32:30</td><td>Microsoft</td><td>4625</td><td>Logon</td></tr></table>	Level	Date and Time	Source	Event ID	Task Categ...	Information	30/06/2026 21:37:21	Microsoft ...	4625	Logon	Information	30/06/2026 21:37:05	Microsoft ...	4625	Logon	Information	30/06/2026 21:32:59	Microsoft ...	4625	Logon	Information	30/06/2026 21:32:47	Microsoft ...	4625	Logon	Information	30/06/2026 21:32:33	Microsoft ...	4625	Logon	Information	30/06/2026 21:32:30	Microsoft	4625	Logon	☒
Level	Date and Time	Source	Event ID	Task Categ...																																		
Information	30/06/2026 21:37:21	Microsoft ...	4625	Logon																																		
Information	30/06/2026 21:37:05	Microsoft ...	4625	Logon																																		
Information	30/06/2026 21:32:59	Microsoft ...	4625	Logon																																		
Information	30/06/2026 21:32:47	Microsoft ...	4625	Logon																																		
Information	30/06/2026 21:32:33	Microsoft ...	4625	Logon																																		
Information	30/06/2026 21:32:30	Microsoft	4625	Logon																																		
6.3 Filter Log: Perubahan Hak Akses	Screenshot properti filter (Event ID terkait hak akses) dan hasil lognya	Custom View Properties Filter XML Logged: Any time Event level: ☒ Critical ☒ Warning ☒ Verbose ☒ Error ☒ Information ☒ By log Event logs: Security ☐ By source Event sources: Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76 4672, 4732, 4670 Task category: Keywords: User: <All Users> Computer(s): <All Computers> Clear OK Cancel	☒																																			

			Number of events: 2,368 <table><tr><th>Level</th><th>Date and Time</th><th>Source</th><th>Event ID</th><th>Task Categ...</th></tr><tr><td>Information</td><td>29/06/2026 22:05:10</td><td>Microsoft ...</td><td>4672</td><td>Special Log...</td></tr><tr><td>Information</td><td>29/06/2026 22:04:17</td><td>Microsoft ...</td><td>4672</td><td>Special Log...</td></tr><tr><td>Information</td><td>29/06/2026 22:04:17</td><td>Microsoft ...</td><td>4672</td><td>Special Log...</td></tr><tr><td>Information</td><td>29/06/2026 22:04:16</td><td>Microsoft ...</td><td>4672</td><td>Special Log...</td></tr><tr><td>Information</td><td>29/06/2026 22:03:47</td><td>Microsoft ...</td><td>4672</td><td>Special Log...</td></tr><tr><td>Information</td><td>29/06/2026 22:03:44</td><td>Microsoft ...</td><td>4672</td><td>Special Log...</td></tr></table>	Level	Date and Time	Source	Event ID	Task Categ...	Information	29/06/2026 22:05:10	Microsoft ...	4672	Special Log...	Information	29/06/2026 22:04:17	Microsoft ...	4672	Special Log...	Information	29/06/2026 22:04:17	Microsoft ...	4672	Special Log...	Information	29/06/2026 22:04:16	Microsoft ...	4672	Special Log...	Information	29/06/2026 22:03:47	Microsoft ...	4672	Special Log...	Information	29/06/2026 22:03:44	Microsoft ...	4672	Special Log...	
Level	Date and Time	Source	Event ID	Task Categ...																																			
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Information	29/06/2026 22:03:47	Microsoft ...	4672	Special Log...																																			
Information	29/06/2026 22:03:44	Microsoft ...	4672	Special Log...																																			
6.4 Filter Log: Modifikasi GPO/Kebijakan an	Screenshot properti filter (Event ID terkait GPO) dan hasil lognya	Custom View Properties FilterXML Logged: Any time Event level: ☒ Critical ☒ Warning ☒ Verbose ☒ Error ☒ Information ☒ By log Event logs: Security ☐ By source Event sources: Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76 4719, 5136 Task category: Keywords: User: <All Users> Computer(s): <All Computers> Clear OKCancel Number of events: 12 <table><tr><th>Level</th><th>Date and Time</th><th>Source</th><th>Event ID</th><th>Task Categ...</th></tr><tr><td>Information</td><td>29/06/2026 22:05:23</td><td>Microsoft ...</td><td>4719</td><td>Audit Polic...</td></tr><tr><td>Information</td><td>29/06/2026 22:05:23</td><td>Microsoft ...</td><td>4719</td><td>Audit Polic...</td></tr><tr><td>Information</td><td>29/06/2026 22:05:23</td><td>Microsoft ...</td><td>4719</td><td>Audit Polic...</td></tr><tr><td>Information</td><td>29/06/2026 22:05:23</td><td>Microsoft ...</td><td>4719</td><td>Audit Polic...</td></tr><tr><td>Information</td><td>29/06/2026 22:05:23</td><td>Microsoft ...</td><td>4719</td><td>Audit Polic...</td></tr><tr><td>Information</td><td>29/06/2026 22:05:23</td><td>Microsoft ...</td><td>4719</td><td>Audit Polic...</td></tr></table>	Level	Date and Time	Source	Event ID	Task Categ...	Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...	Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...	Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...	Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...	Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...	Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...	☒	
Level	Date and Time	Source	Event ID	Task Categ...																																			
Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...																																			
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Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...																																			
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Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...																																			
Information	29/06/2026 22:05:23	Microsoft ...	4719	Audit Polic...																																			

7 Otomatisasi Sistem Windows – 30%				
7.1 Skrip PowerShell Monitoring	Lampiran file kode sumber skrip .ps1 yang anda buat (link diizinkan)	script monitoring user <pre>\$webhookUrl = "https://discordapp.com/api/webhooks/1518620355327950998/p815tk5_lu1pFhJkUdMeYwmJKfJREY7956gfoQZ5rsJTdZ1hblHGhKesUxdK7N8Ysg_s" \$event = Get-WinEvent -FilterHashtable @(LogName='Security'; ID=4625) -MaxEvents 1 -ErrorAction SilentlyContinue if (\$event) { # Ekstrak data dari XML Log agar akurat \$eventXml = [xml]\$event.ToXml() \$eventData = \$eventXml.Event.EventData.Data \$targetUser = (\$eventData Where-Object { \$_.Name -eq "TargetUserName" }).Text \$ipAddress = (\$eventData Where-Object { \$_.Name -eq "IpAddress" }).Text \$time = \$event.TimeCreated.ToString('yyyy-MM-dd HH:mm:ss') # Abaikan peringatan jika yang gagal adalah sistem internal Windows (seperti akun DWM-1) if (\$targetUser -notmatch "^(DWM-)" { \$message = "SECURITY ALERT: Gagal Login (Indikasi Brute Force) ** n" \$message = "-----"n \$message = " **Target Akun**: \$targetUser"n \$message = " **IP Asal**: \$ipAddress"n \$message = " **Waktu**: \$time" \$payload = @{ content = \$message } ConvertTo-Json \$utf8Payload = [System.Text.Encoding]::UTF8.GetBytes(\$payload) Invoke-RestMethod -Uri \$webhookUrl -Method Post -Body \$utf8Payload -ContentType 'application/json; charset=utf-8' } }</pre> script monitoring cpu,ram.memory <pre>File Edit View #URL Webhook Discord \$webhookUrl = "https://discord.com/api/webhooks/1522609469849075743/9hR7HSM4qHGKsGpEqJokR5_1FLm13PX9V_DBYth1ckd05gwb1_FuT91UPApbHg2KVu" #Batas Ambang Alert Kritis (X) \$CPU_Threshold = 50 \$RAM_Threshold = 90 #1.Hitung Penggunaan RAM \$os = Get-CimInstance Win32_OperatingSystem \$totalRam = [Math]::Round(\$os.TotalVisibleMemorySize / 1MB, 2) \$freeRam = [Math]::Round(\$os.FreePhysicalMemory / 1MB, 2) \$usedRam = [Math]::Round(\$totalRam - \$freeRam, 2) \$ramPercent = [Math]::Round((\$usedRam / \$totalRam) * 100, 2) #2.Hitung Penggunaan CPU \$cpuLoad = (Get-CimInstance Win32_Processor).LoadPercentage if (\$null -eq \$cpuLoad) { \$cpuLoad = (Get-Counter "\Processor(_Total)\% Processor Time" -ErrorAction SilentlyContinue).CounterSamples.CookedValue \$cpuLoad = [Math]::Round(\$cpuLoad, 2) } #3.Hitung Penggunaan Disk C: \$disk = Get-CimInstance Win32_LogicalDisk -Filter "DeviceID='C:'" \$totalDisk = [Math]::Round(\$disk.Size / 1GB, 2) \$freeDisk = [Math]::Round(\$disk.FreeSpace / 1GB, 2) \$usedDisk = [Math]::Round(\$totalDisk - \$freeDisk, 2) \$diskPercent = [Math]::Round((\$usedDisk / \$totalDisk) * 100, 2) #4.Ambil 5 Proses Teratas (Menggunakan format string standar tanpa karakter khusus ilegal) \$topProcesses = Get-Process Sort-Object WorkingSet -Descending Select-Object -First 5 ForEach-Object { "-, \$(\$_.Name) : \$(\$([Math]::Round(\$_.WorkingSet / 1MB, 2))) MB" } \$processList = \$topProcesses -join "n" #5.Logika Penentuan Status Alert \$isCritical = \$false \$statusTitle = "WARNING SERVER PERFORMANCE REPORT" \$embedColor = 1447003 if (([double]\$cpuLoad -gt [double]\$CPU_Threshold -or [double]\$ramPercent -gt [double]\$RAM_Threshold) { \$isCritical = \$true \$statusTitle = "ALERT KRITIS: PERFORMA SERVER OVERLOAD!" \$embedColor = 1518832 } #Cek argumen dari Task Scheduler untuk membedakan kiriman berkala atau instan \$forceSend = \$args[0] -eq "force" #Jalankan pengiriman HANYA JIKA statusnya Kritis, ATAU dijalankan terjadwal tiap jam if (\$isCritical -or \$forceSend) { # Buat Payload JSON untuk Discord Embeds \$bodyObject = @{ embeds = @(@{ title = \$statusTitle color = \$embedColor fields = @(@{ name = "CPU Usage"; value = "\$cpuLoad %"; inline = \$true } @{ name = "RAM Usage"; value = "\$ramPercent % (\$usedRam GB / \$totalRam GB)"; inline = \$true } @{ name = "Disk C: Usage"; value = "\$diskPercent % (\$usedDisk GB / \$totalDisk GB)"; inline = \$true }) footer = @{ text = "Dibuat otomatis oleh Monitoring Script" } }) } # Konversi ke string JSON dengan aman \$bodyJson = ConvertTo-Json -InputObject \$bodyObject -Depth 4 # Proses Pengiriman ke Webhook Discord dengan Encoding UTF-8 yang eksplisit Invoke-RestMethod -Uri \$webhookUrl -Method Post -body ([System.Text.Encoding]::UTF8.GetBytes(\$bodyJson)) -ContentType 'application/json; charset=utf-8' }</pre>		
7.2 Integrasi API (Telegram/Discord)	Screenshot baris skrip yang memuat Webhook/To	API : https://discordapp.com/api/webhooks/1518620355327950998/p815tk5_lu1pFhJkUdMeYwmJKfJREY7956gfoQZ5rsJTdZ1hblHGhKesUxdK7N8Ysg_s		

	ken & Chat ID	5iu1xPfhUuDMeYvmjKfJREY7956gfoQZ5rsJTdZlhbNGhKesUxdk7N8Mysg_s	
		#URL Webhook Discord \$webhookUrl = "https://discord.com/api/webhooks/1522609469849075743/9hR7MSMqNGKsGFpEqJokR5_1FLmN3PX9V_DBYtHJckoDSgoWBJ_FuT91UPAbHg2KVu" API: https://discord.com/api/webhooks/1522609469849075743/9hR7MSMqNGKsGFpEqJokR5_1FLmN3PX9V_DBYtHJckoDSgoWBJ_FuT91UPAbHg2KVu	
7.3 Konfigurasi Task Scheduler	Screenshot properti Task Scheduler menggunakan opsi "Create Task"	 <pre>PS C:\PBL202> >> \$action = New-ScheduledTaskAction -Execute "powershell.exe" -Argument "-ExecutionPolicy Bypass -WindowStyle Hidden -file C:\PBL202\Scripts\MonitoringLogin.ps1" >> >> \$trigger = New-CimInstance -CimClass (Get-CimClass -ClassName MSFT_TaskEventTrigger -Namespace Root\Microsoft\Windows\TaskScheduler) -ClientOnly >> \$triggers.Subscription = @' >> <QueryList> >> <Query Id="0" Path="Security"> >> <Select Path="Security">*[System[(EventID=4625)]]</Select> >> </Query> >> </QueryList> >> '@ >> \$trigger.Enabled = \$true >> Register-ScheduledTask -TaskName "Alert Brute Force Discord" -Action \$action -Trigger \$trigger -User "SYSTEM" -RunLevel Highest -Force TaskPath TaskName State ----- \ Alert Brute Force Discord Ready</pre>	☒
7.4 Pengujian Program	Screenshot program saat dieksekusi	 <pre>PS C:\PBL202> Get-ScheduledTask -TaskName "Alert Brute Force Discord" Format-List TaskName, State, Author, Actions, Triggers TaskName : Alert Brute Force Discord State : Ready Author : Actions : {MSFT_TaskExecAction} Triggers : {MSFT_TaskEventTrigger}</pre>	☒
7.5 Pengujian Otomatisasi & Alert	Screenshot notifikasi/pesan alert yang berhasil masuk di Telegram/Discord	hasil monitoring user  hasil monitoring cpu,ram,memory	☒

 **peforma** APP 8:12 PM

ALERT KRITIS: PERFORMA SERVER OVERLOAD!

CPU Usage

12 %

RAM Usage

72.5 % (1.45 GB / 2 GB)

Disk C: Usage

14.76 % (14.63 GB / 99.13 GB)

Top 5 Memory-Consuming Processes

- mysqld : 240.69 MB
- MsMpEng : 221.79 MB
- powershell : 185.68 MB
- dns : 125.46 MB
- powershell : 74.7 MB

Dibuat otomatis oleh Monitoring Script

Hasil Monitoring Login Website

 **Login Web** APP 06/07/2026 20:53

NOTIFIKASI AKTIVITAS LOGIN

Username Attempt

admin

Status

SUKSES

IP Address

192.168.104.222

Jam Akses

2026-07-06 20:53:55 WIB

Request Method / URL

POST -> /login.php

User Agent

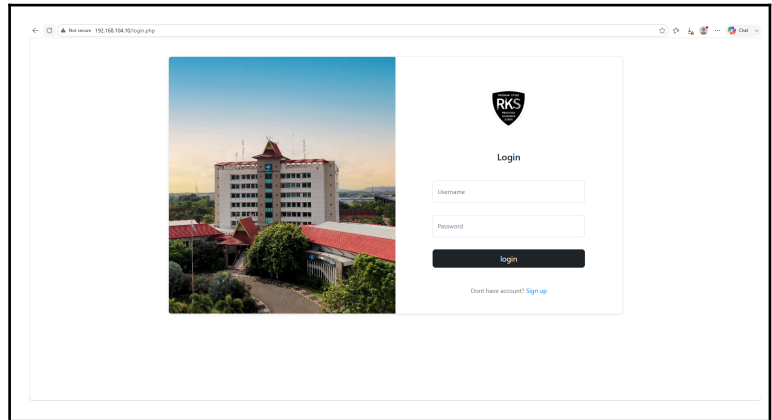
Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0
Safari/537.36 Edg/150.0.0.0

Windows Server 2022 Core Monitor

8 Penerapan Kriptografi

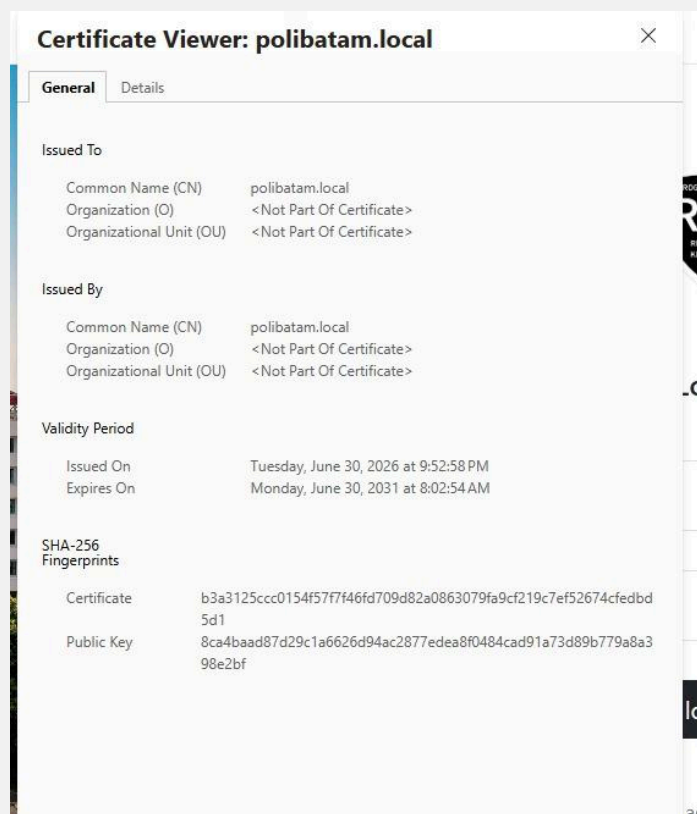
8.1 Implementasi HTTPS (SSL/TLS)

Mengenkripsi Komunikasi antara server dan client serta mencegah terjadinya penyadapan saat mengakses website



8.2 Implementasi Digital Certificate

Dasar mekanisme Public Key infrastructure(PKI) walau masih bersifat self-signed certificate.



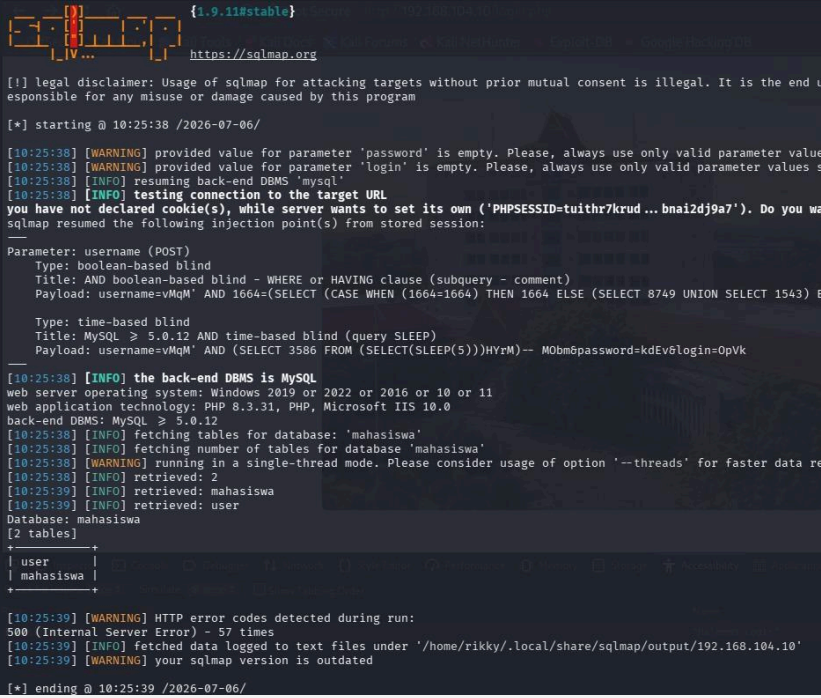
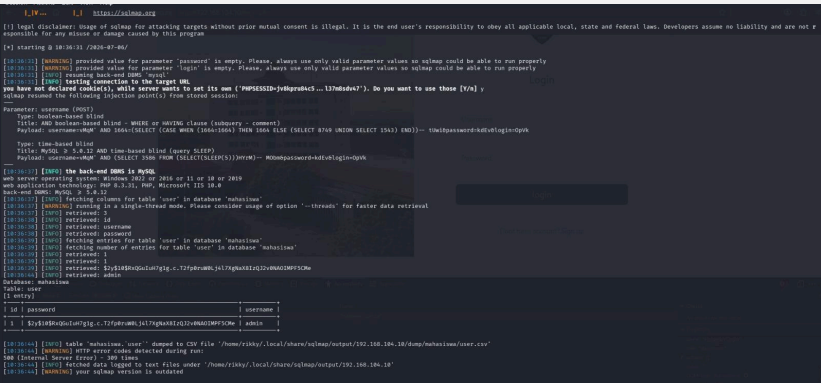
8.3 Implementasi Password Hashing

Mengamankan kredensial user di database agar tidak tersimpan dalam bentuk

```
60
61
62 INSERT INTO `user` (`id`, `username`, `password`) VALUES
63 (1, 'admin', '$2y$10$RxQGuIuH7g1g.c.T2fp0ruw0Lj417XgNaX8IzQJ2v0NAOIMPF5CMe');
64
65
```

		teks biasa (plaintext)	
Kesimpulannya, HTTPS berguna untuk mengamankan komunikasi antara server dengan website. Lalu digital certificate berfungsi sebagai identitas asli dari domain. digital certificate memiliki SHA-256 Fingerprints yang berfungsi sebagai fungsi untuk memastikan keamanan dan keaslian dari website.			

9 Simulasi Serangan SQL Injection Pada Web Server			
9.1 Reconnaissance	Scan port dan service secara detail yang berjalan pada windows server	<pre>➜ nmap -sV -sC -O -p 22,80,135,139,443,445,389,3306,3389, 192.168.104.10 192.168.104.20 192.168.104.30 Starting Nmap 7.95 (https://nmap.org) at 2026-07-06 09:39 EDT Nmap scan report for 192.168.104.10 Host is up (0.0017s latency). PORT STATE SERVICE VERSION 22/tcp filtered ssh 80/tcp open http Microsoft IIS httpd 10.0 _ http-title: Halaman Login _ Requested resource was login.php _ http-server-header: Microsoft-IIS/10.0 _ http-methods: _ Potentially risky methods: TRACE _ http-cookie-flags: _ /: _ PHPSESSID: _ httponly flag not set 135/tcp open mrpc Microsoft Windows RPC 139/tcp open netbios-ssn Microsoft Windows netbios-ssn 389/tcp open ldap Microsoft Windows Active Directory LDAP (Domain: polibatam.local0., Site: Default-First-Site-Name) 443/tcp open ssl/http Microsoft IIS httpd 10.0 _ tls-alpn: _ http/1.1 _ ssl-cert: Subject: commonName=polibatam.local _ Subject Alternative Name: DNS:polibatam.local, DNS:192.168.104.10 _ Not valid before: 2026-06-30T14:52:58 _ Not valid after: 2031-06-30T01:02:34 _ http-title: Halaman Login _ Requested resource was login.php _ http-methods: _ Potentially risky methods: TRACE _ http-server-header: Microsoft-IIS/10.0 _ http-cookie-flags: _ /: _ PHPSESSID: _ httponly flag not set _ ssl-date: TLS randomness does not represent time 445/tcp open microsoft-ds? 3306/tcp filtered mysql 3389/tcp filtered ms-wbt-server NMC Address: 8B40B272D013414 (PCS Systemtechnik/Oracle VirtualBox virtual NIC) Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port Device type: general purpose Running (XOST GUESSING): Microsoft Windows 2022 2016 11 (96%) OS CPE: cpe:/o:microsoft:windows_server_2022 cpe:/o:microsoft:windows_server_2016 cpe:/o:microsoft:windows_11 Aggressive OS guesses: Microsoft Windows Server 2022 (96%), Microsoft Windows Server 2016 (91%), Microsoft Windows 11 21H2 (90%) No exact OS matches for host (test conditions non-ideal). Network Distance: 1 hop Service Info: Host: SERVER-PBL; OS: Windows; CPE: cpe:/o:microsoft:windows</pre>	☒
Target : Windows Server 192.168.104. 10			
Tools : Nmap		Port 80, 443 (IIS), 53 (DNS), 389 (LDAP/AD), 445 (SMB), 3306 (MySQL kalau expose).	

	9.2 Dump Database	Mengambil struktur table user pada Database mahasiswa dengan tools SQL map		☒
	9.3 Hasil SQL Injection	Table user pada database terekspos dari serangan SQL Injection.		

10	Konfigurasi dan Mitigasi Pengerjaan		
	10.1 Network	Konfigurasi Network yang digunakan adalah Bridge Adapter yang membuat VM berada di Subnet yang sama dengan host. agar IP tidak berubah-ubah maka diperlukan settingan IP Static. Hambatan kami pada konfigurasi ini adalah interface network sempat mengalami kegagalan karena kabel LAN yang kami gunakan tercabut sehingga interface network (Ethernet) tidak terbaca oleh VM.	Mitigasi yang kami lakukan pastikan kabel LAN tersambung dan interface network (Ethernet) terbaca.
	10.2 Akses Domain	domain polibatam.local ketika membuka web itu hanya berlaku ke client yang dns nya mengarah ke ad server.	Mitigasi yang kami lakukan adalah jika di akses dari client lain, maka harus membukanya lewat IP.

	10.3 Binding HTTPS (SSL/TLS)	Pada saat implementasi HTTPS di website yang kami gunakan, Domain sempat tidak dikenali oleh web server (IIS). Hal ini menyebabkan munculnya tampilan connection not secure pada website. padahal HTTPS telah berjalan.	Mitigasi yang kami lakukan adalah melakukan Binding atau pengikatan HTTPS pada web server (IIS). Hasilnya Webserver mengenali domain dan tampilan connection not secure tidak muncul lagi.
	10.4 Bcrypt sebagai Algoritma Hashing	Database masih rentan terhadap serangan, dibuktikan dengan simulasi serangan SQL Injection yang berhasil menembus Database user.	Mitigasi yang kami lakukan adalah menambahkan algoritma hashing bcrypt dibanding MD5 dan SHA pada password yang rentan terhadap Rainbow Table Attack. Hal ini sebagai proteksi tambahan jika database bocor, maka penyerang tidak langsung mendapatkan kredensial user.

Note:

- Jobsheet ini akan dikumpulkan sebagai bentuk hasil AAS Praktikum
- Pada kolom sukses beri tanda centang jika implementasi berhasil dilakukan dan bukti tersedia.
- Silahkan menambahkan baris kolom pada tiap kategori jika ada tugas yang anda eksplorasi di luar jobsheet ini